

Solution Requirements

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

This document defines a comprehensive list of requirements that describe exactly what CaptionAI will do and how well it is expected to perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of the system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users should be able to create an account and log in securely to access personalized features such as saved captions and caption history.	High
2	Authorization levels	The system should provide different access permissions. Regular users can generate, save, and download captions, while the administrator can manage users and monitor application activities.	Medium
3	External interfaces	The application should integrate with the Groq API (LLaMA 3.3-70B Versatile Model) to generate AI-powered captions, hashtags, and call-to-action suggestions based on user input.	High
4	Transactions processing	The system should receive the user's prompt, process it through the AI model, generate captions, and display the output without losing user data.	High
5	Reporting	Users should be able to view previously generated captions through the History page and download or copy them whenever required.	Medium
6	Business rules, etc.	The generated captions should match the selected social media platform, preferred tone, and language provided by the user. The system should also generate suitable hashtags and call-to-action suggestions.	High

S.No	Requirement Category	Requirement Description	Priority
7	Compliance to laws or regulations	The application should protect user information and API credentials while following basic data privacy and security practices.	Medium
8	Other	The system should allow users to regenerate captions, copy results instantly, download generated captions, and maintain caption history for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate captions quickly to provide a smooth user experience.	Generate AI responses within 5 seconds under normal conditions.
2	Scalability	The system should support an increasing number of users without affecting performance.	Support at least 100 concurrent users.
3	Security & Data Privacy	User credentials and API communication should be protected using secure methods.	Use secure password storage and HTTPS for data transmission.
4	Reliability & Availability	The application should remain available and operate without unexpected failures.	Maintain 99% uptime during deployment.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use across desktops, tablets, and mobile devices.	Responsive design with user-friendly navigation.
6	Other (Maintainability)	The application should have a modular and well-organized code structure to simplify maintenance and future updates.	Code should be properly documented and easy to modify.